

Equipment	N132051	Fluid volume	16138
Serial No.	04873100151	Temperature rise (°C)	55
Apparatus type	XFMR	Three-Phase	False
Owner	TNC	In service	True
Region	SNR-NS0	In service date	2002-05-01
Substation	Tracy	Tank	MAIN
Designation	TCY KT2A-A	Norms used	TRN_IEEE_INC_345KV
Norms	TRN_IEEE_INC_345KV	Gassing status	1
Fluid type	OIL	DGA result	1→1
Description	AUTO TRANSFORMER	Breathing result	1→1
Manufacturer	HYUNDAI	Moisture result	1→1
Year manufactured	2001	FQ result	1→1
Model	04873T10015	Trace elements result	0→0
MVA ratings	283	Particle analysis result	0→0
Max rated voltage	525	PCB result	0→0
Oil preservation type	CONSERVATOR/DIAPHRAGM	Furan result	0→0
Cooling	ONAN/ONAF/OFAF	Oil test status	UNREVIEWED

Last Analysis Done On 2025-10-20

Last sample date	2025-04-16
Tank last modified date	2025-07-22
Equipment last modified date	2025-04-03
Norms last modified date	2021-06-05
Last analysis date	2025-10-20
Read-only samples	0
Excluded samples	0
Total samples	10

Reliability-based DGA Summary

DGA last date	2025-04-16	Cumulative sev liquid %	0.00
DGA samples	10	Cumulative sev paper %	0.00
Gassing events	1	Gassing status	1
Event fault type	N/A	Normalized HF ratio	0.00
HF %/yr combined	0.00		

Reliability-based DGA Diagnosis

No significant production of fault gas (gassing status 1).

Dissolved Gas Analysis

Lab report number	317589-029	299006-022	281605-005	274139-002	254720-019	
Sample date	2025-04-16	2024-04-12	2023-02-22	2022-09-16	2021-05-28	
Sample temp	35	35	45	50	45	°C
Hydrogen (H ₂)	0	5	2	2	2	µL/L
Methane (CH ₄)	15.0	13.0	9.9	13.1	13.2	µL/L
Ethane (C ₂ H ₆)	1.9	1.7	1.4	1.6	1.5	µL/L
Ethylene (C ₂ H ₄)	0.7	0.5	0.3	0.3	0.4	µL/L
Acetylene (C ₂ H ₂)	0.0	0.0	0.0	0.0	0.0	µL/L
Carbon Monoxide (CO)	771*	727*	711*	804*	820*	µL/L
Carbon Dioxide (CO ₂)	1210	1104	1080	1652	1100	µL/L
Oxygen (O ₂)	960	2000	2630	1450	956	µL/L
Nitrogen (N ₂)	22900	23800	24500	24900	21000	µL/L
Total heat gas	18	15	12	15	15	µL/L
TDCG	789	747	724	821	837	µL/L
Equivalent TCG	2.370	2.138	2.019	2.316	2.789	%
Total partial press	25.9	27.5	27.5	26.9	23.0	% atm
Est. safe handling limit	12.4	12.0	12.3	12.2	12.2	%
Predicted composite fault gas	139	136	130	147	150	µL/L
Acetylene/Hydrogen (C ₂ H ₂ /H ₂)		0.0	0.0	0.0	0.0	
CO ₂ /CO	1.569	1.519	1.519	2.055	1.341	
CO/CO ₂	0.637	0.659	0.658	0.487	0.745	
Oxygen/Nitrogen (O ₂ /N ₂)	0.042	0.084	0.107	0.058	0.046	
ETCG/ESHL	0.191	0.178	0.165	0.189	0.228	
NEI-HC	0.06	0.05	0.04	0.05	0.05	kJ/kL
NEI-CO	5.12	4.78	4.67	5.87	5.19	kJ/kL
DGA retest days	365	365	365	365	365	days
DGA retest date	2026-04-16	2025-04-12	2024-02-22	2023-09-16	2022-05-28	
DGA reference days	369.0	415.0	159.0	476.0	234.0	days
DGA result	1	1	1	1	1	
Gassing status	1	1	1	1	1	
HF %/yr combined	0.00	0.00	0.00	0.00	0.00	

Symbol legend: *: abnormal level

Dissolved Gas Analysis Remarks

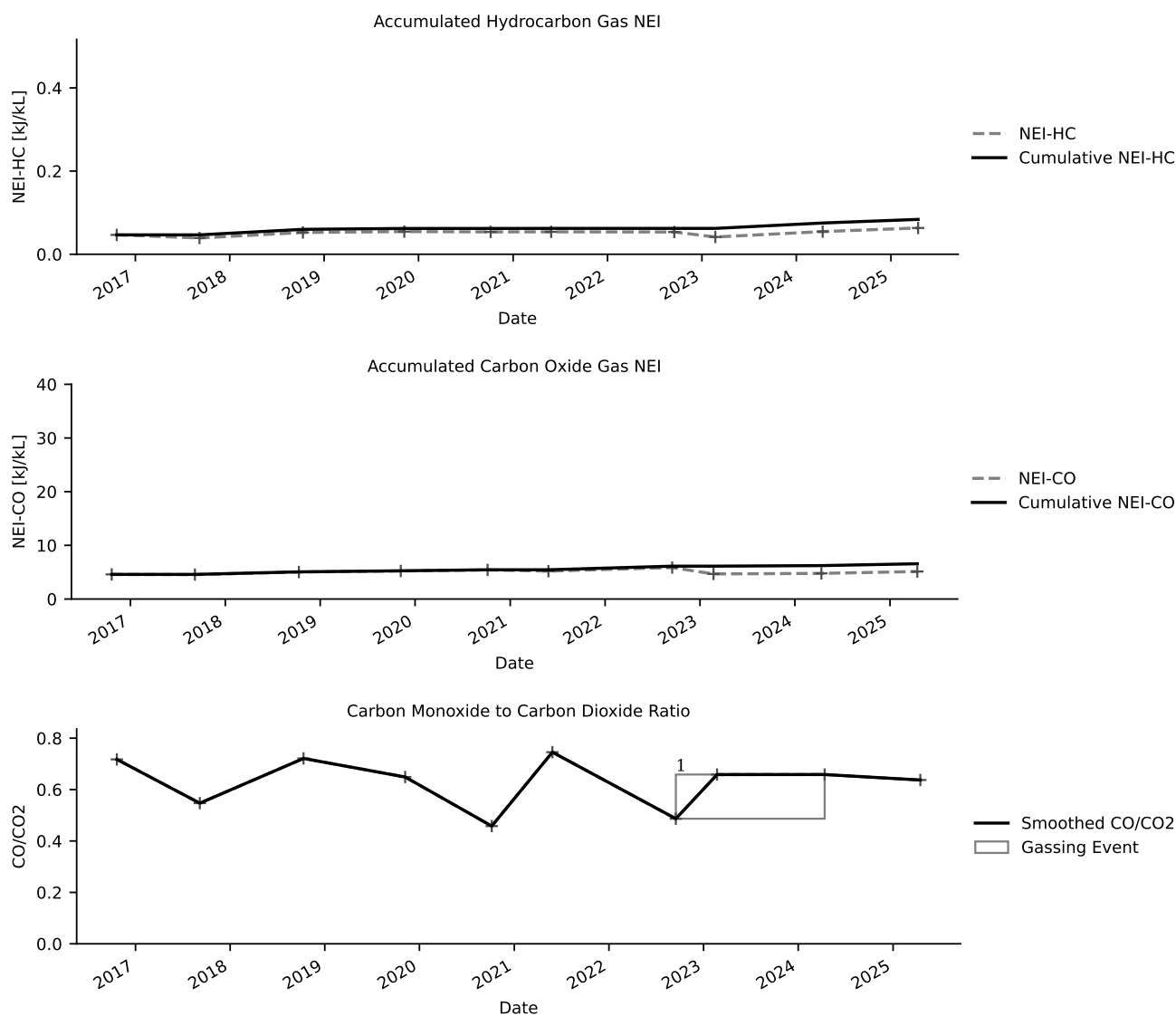
No anomalies.

Dissolved Gas Analysis Summary

Variable name	Value	Units	Description
Carbon Monoxide (CO)	771*	µL/L	Level alert (high 351).

CO*

Gassing Events



ID	Event	Event start	Start value	Inc	Days	Samples	FT	RCOR %	Sev %	HF %/yr
1	R	2022-09-16	0.487	0.172	574	3		35		

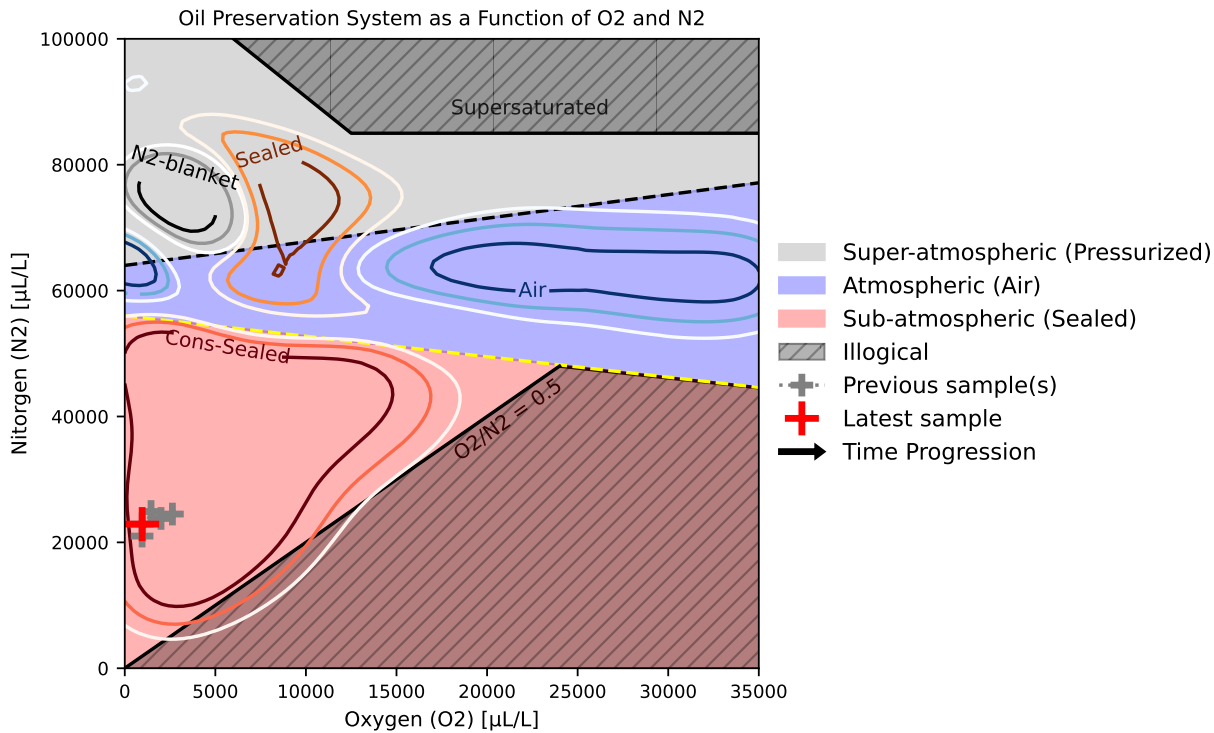
Breathing Status Analysis

Sample date	2025-04-16	2024-04-12	2023-02-22	2022-09-16	2021-05-28	
Oxygen (O2)	960	2000	2630	1450	956	μL/L
Nitrogen (N2)	22900	23800	24500	24900	21000	μL/L
Total partial press	25.9	27.5	27.5	26.9	23.0	% atm
Median O2	1450	1450	1140	1140	1080	
Median N2	23800	23800	23100	21900	21800	
Max deviation O2	1180	1180	1180	1179	1179	
Max deviation N2	3900	3900	3900	3900	3200	
Declared preservation class	CONSERVATOR-SEALED	CONSERVATOR-SEALED	CONSERVATOR-SEALED	CONSERVATOR-SEALED	CONSERVATOR-SEALED	
Predicted preservation class	CONSERVATOR-SEALED	CONSERVATOR-SEALED	CONSERVATOR-SEALED	CONSERVATOR-SEALED	CONSERVATOR-SEALED	
Breathing status result	1	1	1	1	1	

Breathing Status Remarks

Declared oil preservation type is consistent with the data. Appears to be a conservator bladder/diaphragm sealed unit.

O2 & N2 Diagram



Moisture Analysis

Lab report number	317589-029	299006-022	281605-005	274139-002	254720-019
Sample date	2025-04-16	2024-04-12	2023-02-22	2022-09-16	2021-05-28
Sample temp	35	35	45	50	45 °C
Moisture (H2O)	8	5	3	1	2 mg/kg
Relative saturation	8	5	2	1	1 %
Dew point	-18	-26	-34	-50	-40 °C
Moisture result	1	1	1	1	1

Moisture Remarks

The water content of the oil is acceptable.

Fluid Quality

Lab report number	317589-029	299006-022	281605-005	274139-002	254720-019	
Sample date	2025-04-16	2024-04-12	2023-02-22	2022-09-16	2021-05-28	
Sample temp	35	35	45	50	45	°C
Diel brk D1816 (1 mm)	38.0	40.0	38.0	42.0	44.0	kV
Diel breakdown D877	44.0	50.0	48.0	50.0	45.0	kV
PF at 25°C	0.006	0.004	0.005	0.007	0.004	%
PF at 100°C	0.223	0.215	0.183	0.179	0.271	%
Acid number	0.000	0.000	0.000	0.000	0.000	mg KOH/g
Interfacial tension	43.8	45.8	43.8	44.7	44.0	mN/m
Specific Gravity	0.8890	0.8890	0.8889	0.8882	0.8882	
Viscosity	9.130	9.120	9.120	9.130	9.130	cSt
Flash point	148	156	156	152	156	°C
Color	L1.0	0.5	L 0.5	L 0.5	L 0.5	
Oil quality index	0.0	0.0	0.0	0.0	0.0	
PF100/PF25	37.167	53.750	36.600	25.571	67.750	
Visual	C&B	C&B	C&B	C&B	C&B	
Fluid quality retest days	365	365	365	365	365	days
Fluid quality retest date	2026-04-16	2025-04-12	2024-02-22	2023-09-16	2022-05-28	
FQ result	1	1	1	1	1	

Fluid Quality Analysis Remarks

No oil quality problems were detected.

History Graphs and Diagnostic Charts

